Measuring an Equilibrium Constant

Chem 151 Lab

Solutions You Need From Last Week:

- 0.1 M Fe^{3+}
- 4 mM Fe^{3+}
- 6 mM Fe^{3+}
- 8 mM Fe^{3+}

- 0.2 mM SCN^{-}

Today's Equilibrium Reaction

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$$

Initial	**Initial $[Fe^{3+}]$**	**Initial $[SCN^{-}]$**		**0**
Change	**- x**	**- x**		**+ x**
Equilibrium	**$[Fe^{3+}]_0$ - x**	**$[SCN^{-}]_0$ - x**		**x**

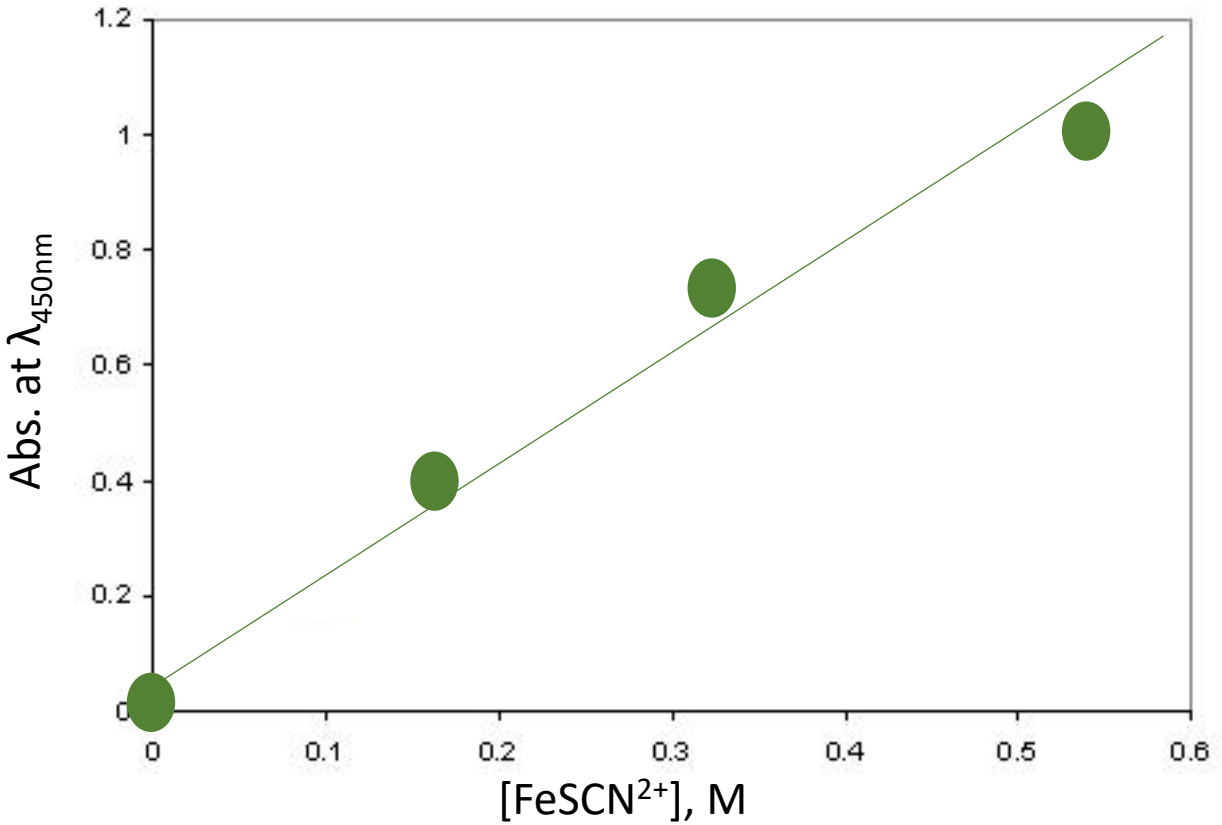
λ_{max} = 450 nm

Measured using Beer's Law

$$K_c = \frac{[FeSCN^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^{-}]_{eq}}$$

Part B: Calibration Standards

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$$



Force Reaction

Then, $[FeSCN^{2+}]_{eq} = [SCN^{-}]_{initial}$

Dilution!

$M_1V_1 = M_2V_2$

Solve for M_2:

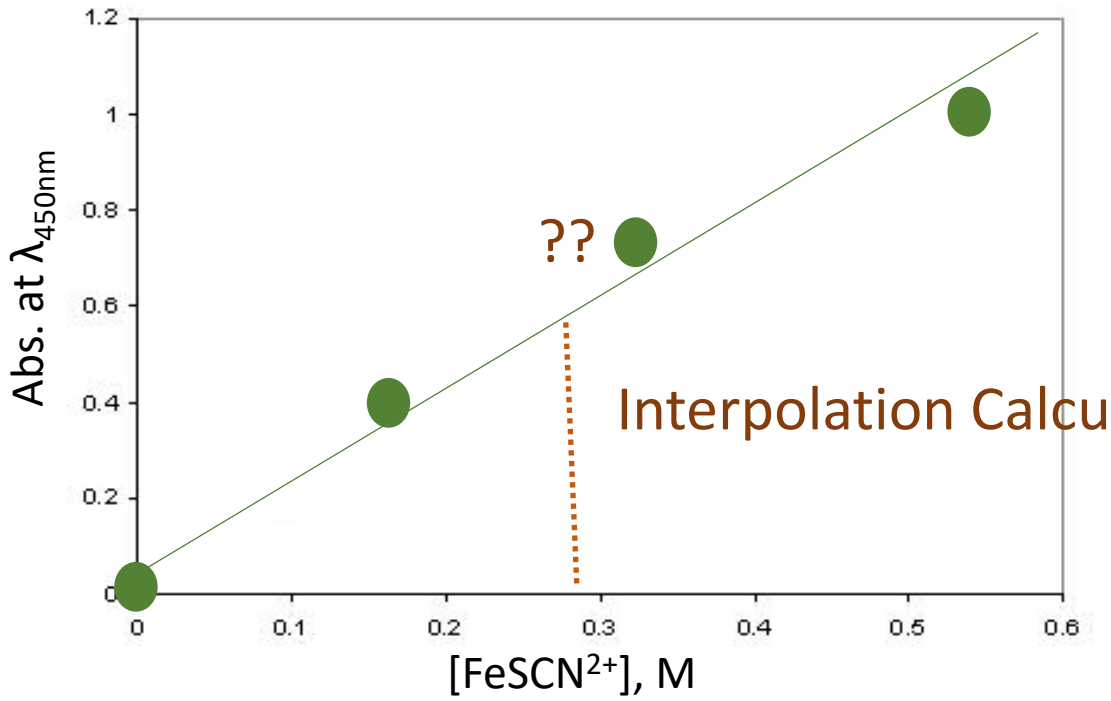
$$M_2 = \frac{M_1V_1}{V_2}$$

30 mL

*Use blank as $[FeSCN^{2+}] = 0$ as a data point!

Solns 1, 2, 3, + blank

Part C: Use the Calibration Plot from Part B



Measure soln 4, 5, 6

Interpolation Calculator = $[FeSCN^{2+}]_{eq}$ in each solution!

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$$

I	_____	_____		0
C	- x	- x		+ x
E	_____ - x	_____ - x		x

Need an ICE table for each soln!

You will calculate 3 K_c values.

$$K_c = \frac{[FeSCN^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^{-}]_{eq}}$$

Also find the average K_c

Logistics for Today's Lab

Have ALL solutions 1, 2, 3, 4, 5, 6 + blank ready and transferred in cuvettes before acquiring the spectrometer to collect data

*Make sure you dispose of all solutions into the proper waste containers in the hood

You can dispose of and clean all Nalgene bottle solutions EXCEPT your large bottle of 0.12 M $S_2O_3^{2-}$